Deployment Document — LavaRápido Vehicular

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1. General Information and Version Control

1.1 Deployment Fact Sheet

1.1.1 System Identification and Version (Release Tag)

Field	Value
System	LavaRápido Vehicular — bookings, payments and operations management for a car wash
Components	6 backend microservices, API Gateway, web application, Android mobile app
Version (Release Tag)	`v1.0.0`, created on 2026-10-23 in 8 repositories (web, api-gateway, security, customer, booking, operations, notification, and infra), where `main` was made equal to `develop` at that moment. payment-service has `v1.0.0` since 2026-10-24: that day its PR #2 `develop` → `main` was merged and the tag was put on that `main` (`2deac9e`), which includes its whole `develop`. **Pending:** the mobile app (PR #1 "Release v1.1.0" open; its `main` dates from 2026-07-06). After the tag, `develop` moved ahead again with the fixes of 2026-10-23 (1.3.8) and the 128 branches of 2026-10-24 (1.3.9), which will ship in `v1.1.0`. payment's PR #2 was merged on 2026-10-24, with the CI and SonarCloud green
Deployed branch	`develop` of 2026-10-24 in the server's 9 repositories, deployed that day (1.3.9). Until `v1.1.0` is released, every update is launched with `BRANCH=develop`: with `main`, the protection in `setup-server.sh` stops without touching anything, because `main` does not contain what is deployed. With `v1.1.0` the server moves to `main`, except payment-service while it stays in `DEVELOP_ONLY`
Mobile app version	**1.1.0** (versionCode **5**), APK published on 2026-10-19 + over-the-air updates (EAS Update)
Repositories	`lavarapido-infra`, `lavarapido-api-gateway`, `lavarapido-security-service`, `lavarapido-customer-service`, `lavarapido-booking-service`, `lavarapido-operation-service`, `lavarapido-notification-service`, `lavarapido-payment-service`, `Front-end-proyecto-web`, `LavadoVehicular-Movil`
Public address	https://lavarapidovehicular.northcentralus.cloudapp.azure.com

1.1.2 Scheduled Date, Time and Maintenance Window

Field	Value
Initial deployment date	October 16, 2026
Maintenance window	02:30 – 05:30 (Colombia time, UTC-5)
Actual duration	~2 h 30 min (provisioning, image builds, HTTPS setup and testing)
User downtime	Not applicable: first deployment (no users in production yet)
Window for future updates	Outside the car wash's business hours (before 07:00 or after 18:00)
Latest update	2026-10-26 around 12:02 (Colombia time): booking-service, payment-service, and the web from `develop`, with payment at booking, booking migration 049 and seed 117 (1.3.11); services healthy and 0 errors. Before that, on 2026-10-24 at 17:12 UTC: notification-service only, with migration 048, which fixes reading `pending_delivery` (1.3.10). Before that, the same day around 15:10: the complete `develop` of the 10 repositories (fixes of October 23 and 24, 1.3.8 and 1.3.9), in two runs over SSH; 11 healthy containers, migrations 044, 045, and seed 113 (booking), 047 and seed 116 (notification), and 022 and 023 (payment) applied, and 0 errors in the logs. The previous update (2026-10-22) is in 1.3.7

1.1.3 Target Environment (VPS / Staging / Production)

Field	Value
Environment	**Academic production** (also used as staging for the team's tests)
Provider	Microsoft Azure (*Azure for Students* subscription)
Type	Linux virtual machine (IaaS), equivalent to a VPS
Region	North Central US
Alternate environment	None: the Hostinger VPS was dropped on 2026-10-21 because the payment was not approved; production stays on Azure only

1.1.4 Roles and Responsibilities Matrix (DevOps, Tech Lead, QA, DBA)

> Responsibilities according to the services each member maintains on GitHub (owner of each repository). The QA and DBA roles have no fixed person: the project leader assigns them and, meanwhile, they are covered as shown.

Role	Owner	Deployment responsibilities
DevOps	Brayan Cabrera	Server, Docker Compose, Caddy/HTTPS, deployment and update scripts
Tech Lead	Brayan Cabrera	Release approval, change review, team coordination
QA	No fixed person (assigned by Brayan Cabrera); meanwhile, each owner tests their part and DevOps runs section 5	Smoke tests and functional validation per role (client, operator, administrator)
DBA	No fixed person (assigned by Brayan Cabrera); meanwhile, DevOps with the scripts of section 3.2	Migrations, backups and database verification
booking-service and customer-service owner	Camilo Guilombo	Changes and verification of their services
payment-service and mobile app owner	Merari Urbano Quintero	Changes and verification of payments and the app

1.2 Document Version History

Version	Date	Author	Description
1.0	2026-10-16	Brayan Cabrera	Initial version: deployment on Azure
1.1	2026-10-16	Brayan Cabrera	Portainer CE installed, real evidence (5.4) and correction of Liquibase schemas and tables
1.2	2026-10-16	Juan Camilo Guilombo	Table count corrected (57) and first administrator created with the environment variables
1.3	2026-10-19	Juan Camilo Guilombo	Update of 2026-10-19 (migration 039 and app 1.1.0), SSH access with a personal key per member, deployment from GitHub Actions, continuous integration in the 10 repositories, and new terminal and web evidence
1.4	2026-10-20	Development team	Topology diagram: customer-service also connects to RabbitMQ (it consumes `security.user_registered` to create the customer profile)
1.5	2026-10-21	Brayan Cabrera	Verified automatic backups (3.2), security headers and CSP (2.2.3 and 5.2.4), health checks and capped logs in every container, `SIMPLE` recovery model, update of 2026-10-21 (1.3.6), code changes of 2026-10-21 pending deployment (1.3.7), and removal of the Hostinger VPS
1.6	2026-10-21	Development team	Evening changes of 2026-10-21 in 1.3.7: confirmed bookings with an operator that nobody registered are no longer closed as "No-show" on their own and go to "To close", migrations 040 (operations) and 041 (notification), `FOLLOW_UP_*` variables, fix for `null` fields in notification-service events, and mobile app adjustments
1.8	2026-10-24	Development team	Deployment of 2026-10-24 with `develop` (1.1.1, 1.1.2, and 1.3.9): two runs, health wait, `:prev` images, `rollback.sh`, and the protection that prevents downgrading; the web on port 8080 (2.2.3); new variables `AUTH_RATE_LIMIT_*`, `TRUSTED_PROXIES`, `ROUTE_*_TIMEOUT`, `START_LATE_LIMIT`, and `INTERNAL_API_KEY` and `BOOKING_SERVICE_URL` in customer-service (3.3.2); rollback with `rollback.sh` and restore with `payment-migrate` (3.2.4 and 3.4); continuous deployment merged into `develop`, without secrets (4.3.4); mobile OTA with `cd.yml`; CI of 2026-10-24 (4.3.3)
1.7	2026-10-23	Development team	Version `v1.0.0` in 8 repositories and what is pending for payment and mobile (1.1.1); deployment of 2026-10-22 (1.3.7); fixes of 2026-10-23 in `develop` for `v1.1.0` (1.3.8); the variables the compose file already passes, `FOLLOW_UP_LOOK_BACK_DAYS` and `ALLOWED_EMAIL_DOMAINS` (3.3.2); a rollback that works (3.4); `setup-server.sh` from `main` (4.3.1); continuous deployment prepared and not yet tested live (4.3.4); QA and DBA owners (1.1.4). Includes the content of Merari Urbano's `fix/A-06`, `fix/A-07`, and `fix/A-08` branches
1.9	2026-10-24	Development team	notification-service update of 2026-10-24 at 17:12 UTC with migration 048 and its result (1.3.10); latest update (1.1.2)
1.10	2026-10-26	Development team	Deployment of payment at booking on 2026-10-26 around 12:02 (Colombia time): booking-service, payment-service, and the web, with their checks (1.3.11); new booking-service variables and `INTERNAL_API_KEY` required in payment-service (3.3.2); latest update (1.1.2)
1.11	2026-10-26	Development team	Versions *v1.1.0* and *v1.1.1*, payment-service from `main`, first run of continuous deployment and of APK *versionCode* 6 with the over-the-air update, and SonarCloud fixes (1.3.12 and 4.3.4).

1.3 Change Summary (Release Notes)

1.3.1 New Features

Platform and deployment

- Full deployment with Docker Compose and a production file (`docker-compose.prod.yml`).
- Automatic HTTPS with Caddy and a Let's Encrypt certificate (automatic renewal).
- Creation of the first administrator when installing on an empty database (`ADMIN_EMAIL` / `ADMIN_PASSWORD`).
- Server installation script (`setup-server.sh`) and one-step update script (`update-server.sh`).
- Android app download from the web (button and QR code) at the fixed address `/app/LavaRapidoVehicular.apk`.

Notifications

- Push notifications in the mobile app (Expo + Firebase) implemented with the Command pattern.
- Delivery through channels (push and email) with failure isolation.

- Notices in the account's language (Spanish, English, French and Portuguese).
- Notice to administrators about upcoming bookings without an operator.
- New app version notice by push and in the inbox.
- The web bell refreshes by itself and shows a floating notice with new notifications.

Bookings and operations

- Automatic operator assignment when a booking is confirmed (RF-008).
- The client can choose the operator among the available ones (RF-010).
- The operator can only start the service from 15 minutes before the booking.
- Vehicle inspection report with phases, findings and photos, shared with the client.
- Daily summary and operator team metrics for the administrator.

Payments and accounts

- Payment history sorted and filterable by date and service (RF-015/016).
- Account lockout for 15 minutes after 5 failed login attempts (RF-004b).
- Business data editable by the administrator and profile address stored in the backend.

Mobile app

- New version notice and over-the-internet updates (EAS Update) without reinstalling.
- Icon with the project logo and the name "LavaRapidoVehicular".

1.3.2 Bug Fixes

- An approved payment was shown as "under verification" on the payments screen (web and mobile).
- The deployed web received 403 on every call because of the services' CORS configuration.
- security-service did not start when an administrator already existed (query outside a transaction).
- The app's operator list did not show a newly created operator until logging out.
- A late 401 after logging out showed the "session expired" notice.
- The new APK announcement could not read the inter-service key because of file permissions.
- Pending reminders kept being sent after the service was started or finished.
- No operator is assigned to bookings whose time has already passed, and cancelled bookings are no longer listed as pending.
- The web requested a non-existent `favicon.ico` (404 error in the console) and used deprecated translation options (console warning).

1.3.3 Infrastructure and Dependency Changes

Change	Detail
Docker images	Java: `eclipse-temurin:21-jdk` (build) and `21-jre` (runtime) · .NET: `dotnet/sdk:8.0` and `aspnet:8.0` · Web: `node:22-alpine` and `nginx:1.27-alpine`
Infrastructure services	`mcr.microsoft.com/mssql/server:2022-latest` (Express), `rabbitmq:4-management`, `caddy:2-alpine`, `liquibase/liquibase:4.29`
New migrations	032 (business data), 033 (inspection report), 034 (login lockout), 035 (address), 036 (registration date), 037 (operator profile), 038 (new-version notice type), and 039 (acceptance of the legal documents, applied on 2026-10-19)
Web dependencies	`qrcode` 1.5.4
Mobile dependencies	`expo-notifications`, `expo-device`, `expo-build-properties`, `expo-updates`, `expo-application`
External services	Expo Push + Firebase Cloud Messaging (push), Gmail SMTP (email), EAS (app builds and updates)
Server operation (2026-10-20 and 21)	Automatic backups with verification (`backup-db.sh`, `verify-backup.sh`, `restore-db.sh`), database in the `SIMPLE` recovery model, health check every 30 s in the 11 containers, logs of each container capped at 3 files of 10 MB, and security headers with a CSP in Caddy

1.3.4 Breaking Changes

- **New mandatory variables** on the server. Without them, Docker Compose does not start:
- `INTERNAL_API_KEY`, `PUBLIC_URL`, `RABBITMQ_USERNAME` and `RABBITMQ_PASSWORD`;
- `RABBITMQ_*` cannot be `guest/guest`.
- **First administrator:** the demo accounts (`admin@gmail.com`, etc.) **only exist in development**. In production the administrator is created with `ADMIN_EMAIL` / `ADMIN_PASSWORD`.
- **Mobile app:** the Android package changed to `com.vehiclecarwash.app`. Previous versions must be uninstalled once; later ones install on top.
- **CORS:** each service only accepts the origin defined in `PUBLIC_URL`.

1.3.5 Update of 2026-10-19

On 2026-10-19 at 03:20 (server time), `update-server.sh` was run. The `develop` changes up to October 17 and 18 were deployed:

- **Database:** migration 039, which stores the date and time each person accepts the Terms and the Data Processing Policy.
- **Backend:** the payment-service health check queries the database, payment-service handles the correlation identifier, the schedule error carries the reason (`reason`), two simultaneous requests no longer fail when creating the same operator, and a bay cannot be created in maintenance.
- **Web:** the payment screen selects by account, the profile option is called "Deactivate my account", Angular is on 21.2.25 and xlsx on 0.20.3, and there are layout fixes for phones.
- **Mobile app:** version 1.1.0 (versionCode 5), published on 2026-10-19 at 05:03, with the version shown in Settings.

The changes pushed afterwards are left for the next update: the error messages with `spring.web.error` in the Java services and the latest web and app fixes.

1.3.6 Update of 2026-10-21

On 2026-10-21 at 05:21 (server time), the images were rebuilt with the `develop` code of October 20 and the new `lavarapido-infra` infrastructure. A database backup was taken first.

- **Infrastructure:** verified automatic backups (3.2); database in the `SIMPLE` recovery model; health check every 30 seconds in the 11 containers, Caddy included (`docker compose ps` shows them `healthy`); logs capped at 3 files of 10 MB per container; security headers and a content security policy (CSP) in Caddy (5.2.4).
- **Backend and web:** the changes left pending on 2026-10-19 (error messages with `spring.web.error` and web fixes) and those of October 19 and 20: the role change of an account was removed, the example business data is cleared so the real business fills it in (booking-service seed 111), and the web opens the detail of each booking and notification.

1.3.7 Changes of 2026-10-21 (Deployed on 2026-10-22)

They were in `develop` since 2026-10-21 and reached the server on 2026-10-22 with `update-server.sh`; seed 112 and migrations 040 and 041 were applied, and backups `LavaRapido-20261022-081543.bak.gz` and `LavaRapido-20261022-173855.bak.gz` were taken that day (the latter when operations-service was deployed again at 12:38):

- **security-service and api-gateway:** the token of a closed session (logout, disabled or deactivated account) stops working; the gateway confirms it with `GET /api/v1/auth/session` and keeps the answer 30 seconds per token.
- **booking-service:** every 5 minutes, a scheduled booking still unconfirmed one hour after its end is cancelled with the system reason `EXPIRED` ("Expired without confirmation"); seed 112 adds that reason, and booking-service applies it on startup. The internal route `POST /internal/v1/bookings/{id}/no-show` sets an expired confirmed booking to "No-show", only when operations-service asks for it (409 `BOOKING_NOT_EXPIRED` if it has not expired yet).
- **operations-service:** manual and automatic assignment use the same lock, and `GET /api/v1/admin/assignments/unassigned` gives the single count of bookings without an operator, now used by the web and the mobile app. Every 5 minutes, `FollowUpJob` sets to "No-show" a confirmed booking that never had an operator and ended more than an hour ago; one that had an operator and that nobody started or finished is never marked on its own: it goes to the administrator's "To close" list (`GET /api/v1/admin/assignments/to-close` and `POST /api/v1/admin/assignments/{bookingId}/attended`), with a single notice to the operator and the administrators. Migration 040 adds `execution.service_execution.follow_up_notice_at`.
- **notification-service:** `OPERATOR_SERVICE_NOT_REGISTERED` and `ADMIN_SERVICE_NOT_REGISTERED` notices (inbox and push, migration 041), and a `null` field in an event payload no longer sends the message to the dead-letter queue (DLQ).
- **payment-service:** receipts of at most 2 MB (`RECEIPT_TOO_LARGE`) and the cash account in the customer list only while it is active.
- **Web:** session in step across tabs, compressed receipt, QR magnifier, reports that count only completed bookings, the roles tab texts, and the "To close" card on the administrator's home.
- **Mobile app:** compressed receipt, QR magnifier, mandatory APK update screen, "To close" card, and count of unassigned bookings with the operations-service rule; the administrator's booking form saves the chosen operator and status, the customer's points bar is full when nothing is left to unlock and does not show "0% OFF", the operator sees "—" when there are no ratings, the roles tab says the 3 roles are fixed, and the new notices open the operator's services screen or the administrator's home. They add no native dependencies, so they arrive with `npm run publish:update`.

The new variables (`SESSION_CHECK_*`, `BOOKING_EXPIRY_*`, and `FOLLOW_UP_*`, nine in total, including `FOLLOW_UP_LOOK_BACK_DAYS`) have defaults, are commented out in `.env.example` and `.env.production.example`, and `docker-`

`compose.yml` passes them to the containers since 2026-10-22 (commit `70b990c` of `lavarapido-infra`): they are only written in `.env.production` to change their value. `FOLLOW_UP_*` needs `INTERNAL_API_KEY`, which is already in `.env.production`.

1.3.8 Version v1.0.0 and Fixes of 2026-10-23 (Deployed on 2026-10-24)

On 2026-10-23 `develop` was merged into `main` and tagged `v1.0.0` in 8 repositories. payment-service and the mobile app do not have that version yet: payment's PR #2 is open, without review and with the SonarCloud quality gate failing on four false positives, and the app has no PR. That same day, after the tag, these fixes from the audit of 2026-10-22 entered `develop` and will ship in `v1.1.0` (CI green in the 10 repositories):

- **api-gateway:** stops publishing `/api/v1/operator/bookings/**` (E-01).
- **booking-service:** the operator routes only accept operations-service with the internal key and apply the 15-minute rule (E-01); the administrator chooses the cancellation reason (A-20); a status change made by the administrator publishes an event for the customer (A-11); loyalty points are frozen on each booking line (A-09, migration 043).
- **operations-service:** sends the internal key to booking-service (E-01), stores the assignment notes (B-01, migration 042), and checks that the operator's account is still active when assigning and rescheduling (A-13, E-12).
- **security-service:** validates the allowed email domains on the server (`ALLOWED_EMAIL_DOMAINS`, `gmail.com` by default, A-01); a token that still carries a role the account no longer has stops working and its sessions are revoked (E-02); it no longer starts in the `dev` profile by default nor keeps demo passwords in the code (E-05).
- **customer-service:** rejects plates longer than 6 characters instead of cutting them (E-16).
- **notification-service:** notifies the customer when the administrator changes the status of their booking (A-11).
- **payment-service:** the percentage discount goes from 1 to 99 % and there is always an amount left to pay (E-07); removing the coupon cancels the redemption on the server (`DELETE /api/v1/loyalty/redemptions/{bookingId}/{promotionId}`) and only one promotion per booking is allowed (migration 021, E-08 and E-09); no default password for its isolated database (E-05).
- **Web:** E-07, E-08, and A-01 as above; reassigning the operator (A-02), notes when assigning (B-01), cancellation reason (A-20), real data in the operator's history and services (A-18), revenue by approval date and the "In service" status (A-17, A-19), message for a deactivated operator account (A-13), and tests for the guard, the interceptor, and several ViewModels (F-02).
- **Mobile app:** E-07, E-08, A-01, and A-13, and the first automated tests with `jest-expo` in the CI (F-01). They are JavaScript changes: they reach the installed APKs with `npm run publish:update`.
- **lavarapido-infra:** `ALLOWED_EMAIL_DOMAINS` and `DEMO_ADMIN_PASSWORD`, `DEMO_OPERATOR_PASSWORD`, and `DEMO_CLIENT_PASSWORD` (`dev` profile only) in the compose file and in `.env.example`, with no real passwords committed (E-05), and `INTERNAL_API_KEY` documented as required for the operator to start and finish services (E-01).

Before deploying them: `INTERNAL_API_KEY` is already in `.env.production`, so change E-01 needs nothing new; the demo accounts do not exist in production; and payment's migration 021 has the precondition `onFail: CONTINUE`: if the database already has a booking with two redeemed coupons, the unique index is not created and there is no warning, so it is worth checking the database before updating. It is also worth checking whether there are accounts with an email that is not @gmail.com (the web and the app no longer let them sign in) and plates longer than 6 characters in `customer.customer_vehicle` (customer-service can no

longer read that vehicle). SEC-01, BIZ-01, BIZ-02, and OPS-01, still open that day, were fixed on 2026-10-24 (1.3.9). Since sign-in accepts any domain (A-01, 1.3.9), the query for accounts with another domain is no longer needed.

1.3.9 Fixes of 2026-10-24 (Deployed That Day)

On 2026-10-24, 128 fix branches of the 10 repositories were merged into `develop` (author brayan, no co-authors, `--no-ff`). Before pushing each repository, its branches were merged onto `develop` in a separate copy, the full suite was run, and it was checked that what was pushed was identical to what was tested. The `develop` CI was green in all 10 (1,192 tests). The main points:

- **api-gateway:** checks the session with any header scheme (SEC-01); 60 requests per minute per IP on `/api/v1/auth/**`, with 429 `RATE_LIMITED` and `Retry-After`, taking the client's real IP only from trusted proxies (SEC-04, SEC-11); 5 s to connect and 60 s to read on each route (SEC-16).
- **security-service:** mandatory consent when signing up (SEC-07); 5 attempts for the recovery code and the current password (423 `TOO_MANY_ATTEMPTS`, SEC-05, SEC-12); even timing for emails that do not exist (SEC-04); passwords of up to 72 bytes (SEC-11).
- **booking-service:** the customer only changes the date and time (BIZ-01); price greater than 0 (BIZ-04, migration 044); optimistic locking on the booking (BIZ-09, migration 045); schedule or bay changes that clash with bookings answer 409 (BIZ-10); `/bookings/**` routes only for customers (BIZ-12); the task that expired bookings was removed (B-07); seed 113 clears the sample data of the business (E-19).
- **customer-service:** does not allow removing a vehicle with active bookings (BIZ-08; it calls booking-service with `BOOKING_SERVICE_URL` and `INTERNAL_API_KEY`); `/vehicles/**` routes only for customers (BIZ-12); reads the long plates already stored (E-16).
- **operations-service:** starting and finishing do not leave the operator blocked (OPS-01, OPS-03, OPS-11); retries and the dead-letter queue `operations-service.booking-confirmed.dlq` (OPS-04; the old `operations-service.events` queue is drained and deleted on startup); a service can be started up to 4 hours after its end time (A-12, `START_LATE_LIMIT`); the operator sees the notes (B-01).
- **notification-service:** pending deliveries with retries (OPS-05, migration 047); the administrator's status changes cancel the reminders and NO_SHOW has its notice (A-11, seed 116); notice to the operator and the administrators when a booking is left without an operator (E-12).
- **payment-service:** the customer's amount and coupon (`GET /loyalty/redemptions/{bookingId}`, BIZ-02); 409 `AMOUNT_CHANGED` when approving an old amount (BIZ-01); percentage coupon with the current total (BIZ-05, migration 022); a single open payment and a unique reference (A-16, migration 023); the administrator's paged queue (E-10); cancelled-booking notice (BIZ-07); inactive cash for the administrator (C-04); SonarCloud green.
- **Web:** the server's amount and a coupon that survives a reload; fresh data without reloading (on entering, on returning to the tab, with each notification, and every 45 s on screens that wait for another person); the 216 error codes translated; read-only permission matrix (H-05); sign-in with any domain and sign-up only with gmail.com (A-01); the operator's notes and start window; nginx without `root` on port **8080** (H-19).
- **Mobile app:** the same as the web for the payment, the refresh (on returning to the screen and with each push), the errors, and the operator, plus the HTTPS URL in the `production` profile (MOV-08), the paged payment queue, and `cd.yml` to

publish the over-the-air update (OTA) when it reaches `main`.

- **lavarapido-infra:** continuous deployment (4.3.4), `:prev` images and `rollback.sh`, image cleanup and backup retention, `restore-db.sh` with a lock and `payment-migrate`, the fix for the user of the backup `cron`, Caddy pointing to `web:8080`, the remote e2e battery (`e2e.yml`), and the protection that prevents downgrading (`setup-server.sh`, commit `d5d6909`).

How it was deployed (2026-10-24, around 15:10). Over SSH, in two consecutive runs:

1. **First run**, with the scripts the server had: it pulled `develop`, backed up the database, rebuilt, and applied the migrations. Since those old scripts do not restart Caddy, the new web (which already listens on 8080) would have stayed behind a Caddy still pointing to `web:80`; that is why the second run was launched right away.
2. **Second run**, with `BRANCH=develop` and the new scripts the first run brought: it kept the running version as `:prev` (8 images), restarted Caddy because the `Caddyfile` changed, waited up to 8 minutes for the 11 containers to be healthy, wrote `backups/deployed-heads`, and installed the `@reboot` task for `boot-update.sh`.

Result: 11 `healthy` containers; migrations and seeds 044, 045, 113, 047, 116, 022, and 023 applied; the `operations-service.booking-confirmed` queue and its `.dlq` created; 8 `:prev` images; 0 errors in the logs, and the web and API routes answering. **Until `v1.1.0` is released**, every manual update is done with `BRANCH=develop bash ~/lavarapido/lavarapido-infra/deploy/update-server.sh`. When the machine is turned on, `boot-update.sh` pulls `main` by default and the protection stops it without touching anything (it is logged in `backups/deploy.log`): the server keeps what it had.

Before releasing `v1.1.0`, in the production database: look for the duplicate open payments that migration 023 rejected (`rejection_reason LIKE 'Pago duplicado%'`) and count the COMPLETED bookings with open lines (they generate false "not started" notices, ACO-35). ACO-33 to ACO-40 remain open in the Corrective Actions document, Table 7, among them the web's unpaginated payment queue (ACO-33).

1.3.10 notification-service Update of 2026-10-24 (17:12 UTC)

While testing production after the 15:10 deployment (1.3.9), notification-service turned out to be unable to read the new table `notification.pending_delivery` (migration 047): `channel` and `language` were `VARCHAR`, Hibernate reads them as `NCHAR` (`use_nationalized_character_data: true`), and the SQL Server driver does not support that conversion. Since the deployment, notification-service had sent no push notification or email at all (welcome, bookings, payments, and reminders); the inbox notices and the password-recovery email (sent by security-service) still worked. 2 deliveries were stuck: 1 push and 1 welcome email.

- **Fix:** migration `048-notification-pending-delivery-nvarchar.sql` (commit `612417a`, merged into `develop` as `3f99df4`): `channel` NVARCHAR(10) NOT NULL and `language` NVARCHAR(5) NULL, with `uq_pending_delivery_channel` and `ck_pending_delivery_channel` re-created; `MARK_RAN` precondition and rollback. It was first tested in `tempdb` with the data kept, and the notification suite passed.
- **Deployment:** notification-service only, on 2026-10-24 at 17:12 UTC.
- **Result:** migration 048 applied; the 2 stuck deliveries went out at 17:13:50 (push) and 17:13:52 (email), and there have been no errors since. It is recorded as AC-87 in the Corrective Actions document.

1.3.11 Payment at Booking: Deployment of 2026-10-26 (Around 12:02, Colombia Time)

So that nobody blocks the schedule with bookings they do not intend to pay for (a question from the preparation of the final presentation and a recommendation of the instructor), the customer chooses how they will pay when booking: with cash they can

have at most 1 active booking (409 `CASH_BOOKING_LIMIT`), and a booking with digital payment without a receipt is cancelled automatically after 30 minutes, or at the start time if that comes first, with the reason `PAYMENT_TIMEOUT` (AC-88 and AC-89 in the Corrective Actions document). What was deployed:

- **booking-service** (`develop` `db156cb`): `paymentMode` in `POST /api/v1/bookings` (400 `INVALID_PAYMENT_MODE`); `paymentMode`, `paymentDeadline`, and `paymentReported` in the booking views; migration `049-booking-payment-mode-deadline.sql` (`payment_mode` NVARCHAR(10), `payment_deadline`, `payment_reported_at`, and the filtered index `ix_booking_payment_pending`; `MARK_RAN` precondition and rollback) and seed `117-seed-payment-timeout-cancellation-reason.sql` (reason `PAYMENT_TIMEOUT`); queue `booking.payment-events` (with its `.dlq`) bound to `carwash.events` with `payment.reported` and `payment.confirmed`; the `PaymentDeadlineJob` task, which asks payment-service before cancelling. Global numbering after the change: structure max. 049 (next 050), seeds max. 117 (next 118); payment stays at 023.
- **payment-service** (`develop` `d94cde5`): `payment.reported` event when a payment with a receipt is reported, and the internal path `GET /internal/v1/payments/bookings/{bookingId}/reported`, which requires `X-Internal-Key` (401 without a key or with a wrong one). Until now payment did not validate any internal key.
- **Web** (`develop` `7f8aca5`): step 4 "How will you pay?", deadline and cash label in the history, payment screen according to the mode, payment method in the administrator's detail, and the Terms and Conditions of 2026-10-26.
- **lavarapido-infra** (`develop` `0f949e5`): the booking variables and payment's internal key in the compose file (3.3.2).
- **Mobile app** (`develop` `acf4e6d`): the same change, by OTA; it has not reached phones yet, because the OTA goes out when merged into `main` (*v1.1.0*) and the `EXPO_TOKEN` is missing.

How it was deployed. With `BRANCH=develop` and the server update script (`update-server.sh`, section 4.3.3), booking-service, payment-service, and the web were deployed, and migration 049 and seed 117 were applied when booking-service started. `docker-compose.prod.yml` now requires `INTERNAL_API_KEY` for payment-service; since it was already in `.env.production` for the other services, payment started healthy.

Checks:

- All services were `healthy` and there were 0 errors in the logs.
- The new columns of `booking.booking` are NVARCHAR and the `PAYMENT_TIMEOUT` reason exists.
- The `booking.payment-events` queue has a consumer.
- Payment's internal path answers 401 without a key and 200 `reported:true` with a key for a booking with an approved payment.
- 0 failures when publishing events.
- The team made a cash booking (#12) and a digital payment booking (#13, with its deadline at 13:06).

The live tests run earlier locally, with the two simulated failures, are in the Test document, section 3.4.

1.3.12 Versions v1.1.0 and v1.1.1 and First Continuous Deployment (2026-10-26, Night, Colombia Time)

- **Versions.** *v1.1.0* in the 10 repositories (the mobile app got its first version) and *v1.1.1* in the web, the mobile app, payment-service, and infra. *v1.1.1* brings the Terms and Conditions with section 6 (bookings, cancellations, and

provision of the service) in the third person and the new section 8 on the inspection report and the vehicle photographs, and the Data Policy with those photographs in the data collected, the purposes, and retention (sections 4, 5, and 12).

- **payment-service from *main*.** payment-service was removed from `DEVELOP_ONLY` in `setup-server.sh` (*lavarapido-infra* `baeebad`, included in *v1.1.1*): the server now pulls `main` in every repository.
- **Continuous deployment.** With the secrets created, merging the "Release v1.1.1" PR into `main` of `lavarapido-infra` launched the **Desplegar en Azure** workflow with `automatic=true` at 22:19, without logging in over SSH. The requests from the other repositories kept coming until 22:48; those that arrived while another one was running were cancelled and served by the next one. Every run that finished ended green: database backup, `:prev`, all services healthy ("Listo: todos los servicios están sanos"), and "Web y API responden 200". The 22:19 run still pulled payment-service from `develop`; since the 22:46 run it pulls it from `main`.
- **Mobile app.** Publishing *v1.1.0* (13:13) failed because the `EXPO_TOKEN` secret was missing and nothing was published. With the secret created, APK *versionCode* 6 was published at 22:39 (`/app/version.json`) and the *v1.1.1* run (22:41) compared the native fingerprint of the commit with the published APK's: since it matched ("La huella coincide: la actualización le llega al APK publicado"), it published the over-the-air update. If the fingerprint does not match, the workflow does not publish and asks for a new APK, so phones do not get an update they cannot use.
- **SonarCloud.** Fixes of 2026-10-26 in api-gateway (a real null access in the authentication path and comparing CIDR networks by content), security, operations, and notification (*@Primary* on the RabbitMQ publishers and the SMTP and Expo senders), the web, and the mobile app (stricter CI and quality warnings).

2. Scope, Topology and Architecture

2.1 Components and Artifacts in Scope

2.1.1 Administration Panel (Portainer CE)

> **Status: installed on 2026-10-16.** Panel to view the containers' status, logs and resource usage and restart them. Deployment and updates are still done with the scripts (section 4.3).

Artifact	Image	Access
Portainer CE	`portainer/portainer-ce:2.21.5` (~11 MB of RAM)	Only through an SSH tunnel (`https://localhost:9443`); listens on `127.0.0.1`, not exposed to the internet
Environment	`local` (socket `/var/run/docker.sock`)	Stack `lavarapido` visible as *Compose · Limited* (created outside Portainer)

2.1.2 Web Frontend Layer (SPA / Nginx Web Server)

Artifact	Technology	Image / container
Web application	**Angular 21** (SPA), i18n in 4 languages	`lavarapido/web:local` → `lavarapido-web`
Web server	nginx 1.27: serves the build and forwards `/api` to the gateway	inside `lavarapido-web`
Reverse proxy and HTTPS	Caddy 2 (Let's Encrypt)	`lavarapido-caddy`
Android mobile app	Expo / React Native, APK built with EAS	Served at `/app/LavaRapidoVehicular.apk`

2.1.3 Backend API Layer (Java Spring Boot / C# .NET Core)

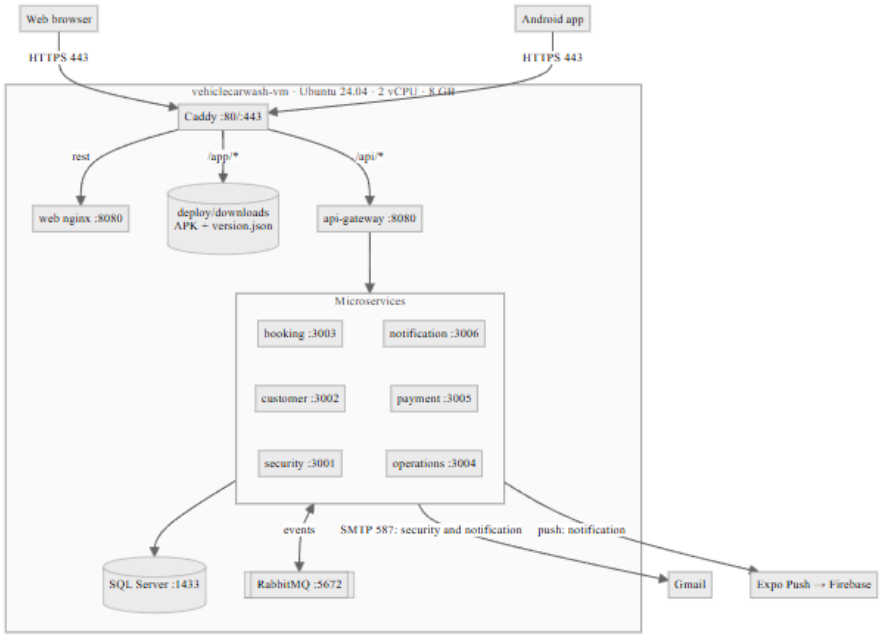
Service	Technology	Internal port	Responsibility
api-gateway	Spring Cloud Gateway (Java 21)	8080	Single entry point for `/api/v1`
security-service	Spring Boot (Java 21)	3001	Accounts, login, JWT, roles, preferences
customer-service	Spring Boot (Java 21)	3002	Clients and vehicles
booking-service	Spring Boot (Java 21)	3003	Catalog, schedule, bookings
operations-service	Spring Boot (Java 21)	3004	Operators, assignment, execution, ratings, inspection
payment-service	ASP.NET Core 8 (C#)	3005	Payments, receipts, loyalty and promotions
notification-service	Spring Boot (Java 21)	3006	Inbox, email, push, reminders
rabbitmq	RabbitMQ 4	5672	Events between services (exchange `carwash.events`)

2.1.4 Persistence Layer (SQL Server)

Artifact	Detail
Engine	**Microsoft SQL Server 2022, Express edition** (container `lavarapido-sqlserver`)
Database	`LavaRapido`: 9 schemas and 57 tables — `security` (11), `customer` (3), `booking` (10), `catalog` (3), `execution` (11), `notification` (6), `payment` (7), `promotion` (5), `audit` (1)
Migrations	**Liquibase**: the Java services apply them on startup; payment-service applies them with the `payment-migrate` container. Each service keeps its history in `dbo.DATABASECHANGELOG_<SERVICE>`
Persistent volumes	`sqlserver-data` (database), `rabbitmq-data` (queues), `caddy-data` (certificates)
Backups	`~/lavarapido/backups/` on the VM disk, outside Docker: the last 14 and the `backup.log` record (3.2)

2.2 Network and Infrastructure Architecture Diagram

2.2.1 VPS Server Topology



2.2.2 Internal Container Network (Docker Bridge Network)

- All containers are on the bridge network *`lavarapido_default`*, created by Docker Compose.

- They call each other by **service name**: `sqlserver`, `rabbitmq`, `security-service`, `api-gateway`, `web`, etc. Docker's internal DNS is `127.0.0.11`.
- The `/internal/**` routes (calls between services) are protected with the `X-Internal-Key` header (`INTERNAL_API_KEY`). **The gateway does not publish them.**
- No port of the internal network is published on the machine, except Caddy's.

2.2.3 Routing and Reverse Proxy With TLS/HTTPS Certificates

Public route	Destination	Notes
`https://<domain>/api/*`	`api-gateway:8080`	REST API v1 for the web and the app
`https://<domain>/app/*`	`deploy/downloads` folder	APK and `version.json` (no cache)
`https://<domain>/*`	`web:8080`	Angular SPA (routes → `index.html`). Since 2026-10-24 the web runs `nginx-unprivileged`, without `root`, on port 8080; its health check also uses 8080
`http://<domain>/*`	308 redirect to HTTPS	Automatic (Caddy)

- **Certificate:** Let's Encrypt, issued and renewed automatically by Caddy (expires on 2027-01-14 and is renewed before). It is stored in the `caddy-data` volume.
- **Domain:** Azure DNS label on the public IP: `lavarapidovehicular.northcentralus.cloudapp.azure.com`.
- **Security headers:** since 2026-10-20 Caddy adds HSTS, `nosniff`, `X-Frame-Options`, `Referrer-Policy`, and `Permissions-Policy` to every response, removes the `Server` header, and sends a content security policy (CSP) with the web (5.2.4).
- **Configuration:** `lavarapido-infra/deploy/Caddyfile`.

2.3 Environments, IP Addressing and Ports Matrix

Environment	Host	IP / URL	Use
Development	Each member's PC	`localhost` (web 4200, mobile 8081, gateway 8080)	Programming and local testing (Mailpit catches the emails)
Academic production	Azure VM `vehiclecarwash-vm`	Static IP `64.236.160.129` · https://lavarapidovehicular.northcentralus.cloudapp.azure.com	Presentation, team testing, installed app

Port	Service	Exposed to the internet?
22	SSH (key only)	Yes (administration)
80 / 443	Caddy	Yes
8080	api-gateway	No (internal network only)
3001–3006	Microservices	No
1433	SQL Server	No
5672 / 15672	RabbitMQ / panel	No
9443	Portainer CE	No (only `127.0.0.1`, through an SSH tunnel)

3. Prerequisites and Preparation (Pre-Deployment)

3.1 Operating System and VPS Prerequisites Checklist

3.1.1 Available Hardware Resources (CPU, RAM, Disk)

Resource	Minimum required	Available (Azure B2as_v2)	Measured in operation	✓
CPU	2 x86-64 vCPU (SQL Server does not run on ARM)	2 AMD x64 vCPU	Peaks only while building images	✅
RAM	8 GB	7.7 GB + 4 GB of swap	**3.3 GB** used with the 12 containers, Portainer included (2026-10-19)	✅
Disk	30 GB	64 GB standard SSD	26 GB used of 61 GB (42%), mostly images and build cache (2026-10-19)	✅
Operating system	Ubuntu 24.04 or 26.04 LTS	Ubuntu 24.04.4 LTS	—	✅
Network	Ports 22, 80 and 443 open	Azure security group	Verified from the internet	✅

3.1.2 Docker Engine Status and Local Socket

```
docker --version           # Docker version 29.x
docker compose version     # Docker Compose v5.x
sudo systemctl is-enabled docker   # enabled  → starts with the machine
sudo systemctl is-active docker    # active
ls -l /var/run/docker.sock         # local engine socket
```

`deploy/setup-server.sh` installs Docker (official `get.docker.com` script) if missing, enables it on boot and creates 4 GB of swap.

3.2 Backup Strategy and Execution

> **Status: automated since 2026-10-21.** Backups, their check and restoring were tested in production (see 3.2.4).

3.2.1 Logical Database Backup (Dump / Snapshot)

`deploy/backup-db.sh` takes a full backup of `LavaRapido` (`BACKUP DATABASE … WITH CHECKSUM`), checks that it can be restored (`RESTORE VERIFYONLY`), copies it out of the container and compresses it with gzip, because SQL Server Express does not compress its backups. It is stored as `~/lavarapido/backups/LavaRapido-YYYYMMDD-HHMMSS.bak.gz`, with `600` permissions, and the **last 14** are kept.

When it runs	Triggered by
Before every update	`update-server.sh` (if the backup fails, **nothing is updated**)
5 minutes after the VM starts	`azureuser` crontab (`@reboot`)
Every day at 23:00 UTC (6 p.m. in Colombia), if the VM is on	`azureuser` crontab
Manually	`bash ~/lavarapido/lavarapido-infra/deploy/backup-db.sh`

The VM is started on demand; that is why, besides the daily job, a backup is taken at start-up and before updating. The database uses the *`SIMPLE`* recovery model (set by `db-init`): with full backups no transaction log backups are needed, and the log does not grow without limit.

Measured on 2026-10-21: the database takes 72 MB and the compressed backup **1.4 MB**; it takes **under 2 seconds**.

Infrastructure-level complement: **disk snapshot** of the VM from the Azure portal (Disks → Create snapshot). Backups live on the VM's own disk: they protect against a human error or a bad migration, not against losing the VM. To keep a copy elsewhere, download them to a PC:

```
scp -i ~/.ssh/lavarapido_azure "azureuser@64.236.160.129:~/lavarapido/backups/*.bak.gz" .
```

3.2.2 Backup of Docker Persistent Volumes

```
mkdir -p ~/backups
for v in lavarapido_sqlserver-data lavarapido_rabbitmq-data lavarapido_caddy-data; do
  sudo docker run --rm -v "$v":/data:ro -v ~/backups:/backup alpine \
    tar czf "/backup/${v}_$(date +%F).tar.gz" -C /data .
done
```

> For a consistent backup of `sqlserver-data`, the `BACKUP DATABASE` from 3.2.1 is preferred, because it can be done while the database is running.

3.2.3 Backup of Stack Configurations and Previous Variables

```
cd ~/lavarapido/lavarapido-infra
cp .env.production ~/backups/env.production_$(date +%F)       # contains secrets: chmod 600
chmod 600 ~/backups/env.production_*
git -C ~/lavarapido/lavarapido-infra rev-parse HEAD > ~/backups/infra-commit_$(date +%F).txt
```

The stack configuration (`docker-compose.yml`, `docker-compose.prod.yml`, `deploy/Caddyfile`) is versioned on GitHub (`lavarapido-infra`). To go back, a `git checkout <commit>` and a new `up -d` are enough.

3.2.4 Automatic Check and Restore

Automatic check. After every scheduled backup, `deploy/verify-backup.sh` runs: it restores the latest backup under another name (`LavaRapido_verificacion`), checks that it has the tables of the real database and can be read, and drops it. This proves every day that the backups work, **without touching the real database**.

Restore. It is never automatic: going back to a backup erases what happened afterwards, so a person always decides it. Since 2026-10-24 `restore-db.sh` takes the same lock as the deployment, requires a terminal, shows the backup's date and the steps before asking for confirmation, runs `payment-migrate` at the end, and never leaves the services stopped. The `LavaRapido-pre-update-*` backups that each update makes are kept separately (the last 10), besides the 14 regular ones. *Open risk (ACO-37):* the safety backup it makes before restoring applies the 14-file retention and could delete the file about to be restored if it is the oldest one; it is better to restore a recent backup or copy it to another folder first.

```
bash ~/lavarapido/lavarapido-infra/deploy/restore-db.sh <file.bak.gz> --as LavaRapido_prueba   # inspect a backup
without touching the database
bash ~/lavarapido/lavarapido-infra/deploy/restore-db.sh <file.bak.gz>     # roll the database back to that backup
```

In the second case the script asks to type `RESTAURAR`, first backs up the current database, stops the 7 services, restores and starts them again (2 to 3 minutes of downtime). If the restore fails, the database stays open as it was and the services start again.

Log. Every backup, check and restore, including cancelled or failed ones, is written to `~/lavarapido/backups/backup.log`:

```
2026-10-21 06:11:16 UTC | RESTAURACIÓN | CANCELADA | LavaRapido-20261021-061111.bak.gz -> LavaRapido | azureuser
2026-10-21 06:12:05 UTC | RESPALDO | OK | LavaRapido-20261021-061204.bak.gz (1.4M) | azureuser
2026-10-21 06:12:08 UTC | VERIFICACIÓN | OK | LavaRapido-20261021-061204.bak.gz: 69 tablas, 4 usuarios, 2 reservas |
azureuser
```

Read it with `tail -20 ~/lavarapido/backups/backup.log`; only problems with `grep FALLÓ ~/lavarapido/backups/backup.log`.

Tests run in production (2026-10-21):

Test	Result
Manual backup and backup before updating	✅ 1.4 MB, verified
Restore under another name and compare with the real database	✅ Same records in users, bookings, payments, notifications and operators; 69 tables
Restore over the real database	✅ Identical data; the 11 containers went back to `healthy`; database `MULTI_USER / ONLINE / SIMPLE`
Check with a deliberately corrupted backup	✅ Detected and logged as `FALLÓ`, leaving no test database or temporary files
Cancelled restore	✅ Logged as `CANCELADA`

3.3 Environment Variables and Secrets Management

- Secrets live **only** in `~/lavarapido/lavarapido-infra/.env.production` (permissions `600`), which is in `.gitignore`.
- The template without values is `.env.production.example`.
- **Key generation:** the keys were generated **on the server** with `openssl rand -base64 48` and were never copied or displayed.
- **First administrator:** the owner wrote `ADMIN_EMAIL` and `ADMIN_PASSWORD` directly in `.env.production` on the server; security-service created the account at startup because there was no administrator yet (`AdministratorBootstrapRunner`). The password is not in any repository.

3.3.1 Database Connection Parameters

Variable	Description
`DB_PASSWORD`	`sa` password (8+ characters, uppercase, lowercase, number and symbol)
`DB_NAME`	`LavaRapido`
`DB_USERNAME`	`sa` *(pending improvement: a user with limited permissions)*
`DB_URL` (compose)	`jdbc:sqlserver://sqlserver:1433;databaseName=LavaRapido;encrypt=true;trustServerCertificate=true`

3.3.2 Token Signing Keys (JWT Secret Keys)

Variable	Description
`JWT_SECRET`	HS256 key shared by the services that validate tokens (≥ 32 bytes). Tokens last **1 hour**
`INTERNAL_API_KEY`	Key for the `/internal/**` routes between services (`X-Internal-Key` header)
`RABBITMQ_USERNAME` / `RABBITMQ_PASSWORD`	Dedicated RabbitMQ user (not `guest`)
`MAIL_USERNAME` / `MAIL_PASSWORD` / `MAIL_FROM`	Gmail account with an app password
`ADMIN_EMAIL` / `ADMIN_PASSWORD`	First administrator (only used if the database has none)
`PUBLIC_URL` / `SITE_ADDRESS`	Public address (CORS, email links) and Caddy domain
`PUSH_ENABLED` / `MESSAGING_ENABLED`	`true`: push with Expo and events with RabbitMQ

The variables of 2026-10-21 are not required in `.env.production`: `SESSION_CHECK_ENABLED`, `SESSION_CHECK_CACHE_TTL`, and `SESSION_CHECK_TIMEOUT` (gateway session check: on, 30 s, and 3 s), `BOOKING_EXPIRY_CHECK_ENABLED` and `BOOKING_EXPIRY_CHECK_INTERVAL` (cancellation of scheduled bookings that expired without confirmation: on and every 5 minutes), `BOOKING_EXPIRY_GRACE_MINUTES` (60-minute grace, used by booking-service and operations-service), and `FOLLOW_UP_ENABLED`, `FOLLOW_UP_INTERVAL`, and `FOLLOW_UP_LOOK_BACK_DAYS` (follow-up of bookings that ended and nobody closed: on, every 5 minutes, and looking back 28 days, because the machine can stay off for weeks) keep their defaults if they are not written; `docker-compose.yml` does pass them to the containers, so they can be changed in `.env.production`. The same applies to `ALLOWED_EMAIL_DOMAINS` (email domains security-service accepts at sign-up, when creating users, and when changing the email, `gmail.com` by default; in production since 2026-10-24).

Variables of 2026-10-24, all with a default value in the service; the compose file does not pass them, so to change them they must be added to the service's `environment` block in `docker-compose.yml`:

Variable	Service	Default value	Purpose
`AUTH_RATE_LIMIT_ENABLED`, `AUTH_RATE_LIMIT_REQUESTS`, `AUTH_RATE_LIMIT_WINDOW`	api-gateway	`true`, `60`, `1m`	Request limit per IP on `/api/v1/auth/**` (429 `RATE_LIMITED`). If the gateway log shows every request coming from a `172.x` IP, all users share the same bucket: check how `X-Forwarded-For` arrives
`TRUSTED_PROXIES`	api-gateway	Private and local networks (`127.0.0.0/8`, `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`, `::1`, `fc00::/7`)	Only the `X-Forwarded-For` coming from these networks (Caddy) is trusted
`ROUTE_CONNECT_TIMEOUT`, `ROUTE_READ_TIMEOUT`	api-gateway	`5s`, `60s`	Time to connect and to read the answer of each service; if exceeded, it answers 500
`START_LATE_LIMIT`	operations-service	`PT4H`	How long after the end time the operator can still start (409 `SERVICE_TOO_LATE`)

In addition, since 2026-10-24 **customer-service** receives `BOOKING_SERVICE_URL` (fixed in the compose file: `http://booking-service:3003`, the service root, without `/api/v1`) and `INTERNAL_API_KEY` (the same as the others; in production it is mandatory and, without it, the compose file does not start), to ask booking-service whether a vehicle has active bookings before removing it. Optional: `BOOKING_SERVICE_TIMEOUT` (3 s). In booking-service, `BOOKING_EXPIRY_CHECK_ENABLED` and `BOOKING_EXPIRY_CHECK_INTERVAL` no longer exist (the task was removed, B-07); the compose file still passes them with no effect (ACO-40). `ADMIN_DOCUMENT` and `ADMIN_PHONE` complete the ID number and mobile phone of the first administrator, next to `ADMIN_EMAIL` and `ADMIN_PASSWORD`. The `DEMO_*_PASSWORD` variables are not used in production: they only exist in the `dev` profile.

Variables of 2026-10-26 (payment at booking). The four booking-service variables have defaults, are commented out in `.env.example`, and the compose file already passes them to the container, so they can be changed in `.env.production`:

Variable	Service	Default value	Purpose
`BOOKING_CASH_ACTIVE_LIMIT`	booking-service	`1`	Maximum active cash bookings per customer (409 `CASH_BOOKING_LIMIT`)
`BOOKING_PAYMENT_WINDOW`	booking-service	`30m`	Deadline to upload the receipt of a booking with digital payment (or the start time, if that comes first)
`BOOKING_PAYMENT_EXPIRY_ENABLED`	booking-service	`true`	Turns on `PaymentDeadlineJob`, which cancels overdue digital bookings with `PAYMENT_TIMEOUT`
`BOOKING_PAYMENT_EXPIRY_INTERVAL`	booking-service	`60s`	How often that task runs
`PAYMENT_SERVICE_URL`	booking-service	`http://payment-service:3005` (fixed in the compose file)	Root of payment-service for the internal query before cancelling
`INTERNAL_API_KEY`	booking-service and payment-service	No default	The same as the other services. Since 2026-10-26 **payment-service also needs it**: `docker-compose.prod.yml` requires it (the compose file does not start without it) and, without it, payment's internal path stays closed and booking cancels no booking for lack of payment

3.3.3 Memory and Runtime Limits (JVM -Xmx, ASPNETCORE_ENVIRONMENT)

Container	Limit (`mem_limit`)	Runtime
5 Java services + gateway	512 MB each	`java -XX:MaxRAMPercentage=75` → maximum heap ≈ 384 MB · `prod` profile (no Swagger)
payment-service	384 MB	`ASPNETCORE_ENVIRONMENT=Production`
SQL Server	`MSSQL_MEMORY_LIMIT_MB=1536`	`MSSQL_PID=Express`
web / caddy	64 MB each	nginx / Caddy
All	`restart: unless-stopped`	They restart by themselves on failure and when the machine boots
All (since 2026-10-20)	Logs: 3 files of 10 MB (`json-file`) · health check every 30 s	The Java services get 150 s to start before failures count. Docker does not restart an `unhealthy` container by itself: the status shows in `docker compose ps`, Portainer, and `update-server.sh`

3.4 Communication Protocol and Maintenance Window

1. **Advance notice (24 h):** the DevOps informs the team through the project group about the date, the window and the changes (release notes).
2. **Freeze:** nothing is merged into the branch being deployed during the window. With continuous deployment (4.3.4) each merge into `main` deploys on its own and recreates containers for 1 or 2 minutes, so `main` is also frozen on the day of the presentation (2026-10-27).
3. **Start:** "deployment started" is announced. `update-server.sh` backs up the database before updating (section 3.2).
4. **Execution:** runbook in section 4.
5. **Verification:** smoke tests in section 5. QA confirms each role.
6. **Closing:** "deployment finished" is announced with the version and the evidence. If it fails, **rollback**. `update-server.sh` pulls the tip of each repository's branch again (it calls `setup-server.sh`), so a `git checkout` made by hand on the server is lost on the next update. Since 2026-10-24 `setup-server.sh` never downgrades: if what is deployed is not contained in the requested branch, it stops without touching anything (`ALLOW_DOWNGRADE=1` allows it on purpose). Therefore:
7. **On GitHub (the normal way):** revert the change on the branch being deployed (`git revert` of the commit, or `git revert -m 1` of the merge, in a `fix/…` branch with its PR) and deploy again: with continuous deployment, merging the revert into `main` is enough.
8. **Emergency on the server:** before rebuilding, `update-server.sh` keeps the running version as `:prev` on its own (if the last update failed, it does not overwrite it). To go back: `ssh -t` into the server and `bash ~/lavarapido/lavarapido-infra/deploy/rollback.sh` (all) or `rollback.sh web` (some). It shows the date of each service's previous version, asks to type `VOLVER`, waits if another update is running (same lock), recreates those containers without building, waits up to 8 minutes for them to be healthy, and logs it in `backups/deploy.log`. It does not touch the database or the repositories: then revert on GitHub. *Open risk (ACO-37):* `rollback.sh` uses the current compose file and `Caddyfile`; if a contract changed between the two versions (like the web's port 80 → 8080), going back may leave that service down, and `update-server.sh` run by hand does not take the lock.
9. **Data:** if the migration damaged data, restore the `LavaRapido-pre-update-…` backup taken before the update with `restore-db.sh` (section 3.2.4); it returns the **whole** database to that moment and, since 2026-10-24, runs `payment-`

`migrate` at the end.

10. **Standard window:** outside the car wash's business hours (before 07:00 or after 18:00, Colombia time).

4. Step-by-Step Deployment Procedure (Runbook)

4.1 Verification and Access to the Base Platform (Portainer CE)

Server access:

```
ssh -i ~/.ssh/lavarapido_azure azureuser@64.236.160.129
```

Access of each member with their own key: each person creates their key on their computer and only shares the public part (`.pub`). The private one is never sent or pushed to a repository.

```
ssh-keygen -t ed25519 -f $HOME\.ssh\lavarapido_azure      # on the person's PC
Get-Content $HOME\.ssh\lavarapido_azure.pub               # this line is the one to share
```

The DevOps owner adds it once, from the portal (*vehiclecarwash-vm → Reset password → Reset SSH public key*, user `azureuser`) or from the server with `echo "<public key>" >> ~/.ssh/authorized_keys`. As of 2026-10-19, the keys with access are the DevOps owner's, Juan Camilo's, and GitHub Actions' (restricted, see 4.3.3). To remove an access, delete its line from `authorized_keys`.

Portainer CE installation *(done on 2026-10-16; listens only inside the server)*:

```
sudo docker volume create portainer_data
sudo docker run -d --name portainer --restart unless-stopped \
  -p 127.0.0.1:9443:9443 \
  -v /var/run/docker.sock:/var/run/docker.sock \
  -v portainer_data:/data portainer/portainer-ce:2.21.5
```

Access from the administrator's PC (SSH tunnel, without opening ports):

```
ssh -i ~/.ssh/lavarapido_azure -L 9443:localhost:9443 azureuser@64.236.160.129
# then open https://localhost:9443 → user "admin"
```

- **Password of the `admin` user:** it was generated during installation and is on the server, in `~/portainer-admin-password` (permissions `600`). It is read with `cat ~/portainer-admin-password` and can be changed from Portainer (*My account*).
- **`local` environment:** registered on the Docker socket. `lavarapido` appears under *Containers* and *Stacks*.

4.2 Database Deployment and Migration

4.2.1 Running DDL/DML Scripts or Migration Tools (Liquibase)

Migrations are **automatic**:

1. The `db-init` container creates the `LavaRapido` database if it does not exist and sets it to the `SIMPLE` recovery model.
2. The `payment-migrate` container (Liquibase 4.29) applies the payment-service changelogs (`db/changelog/db.changelog-master.yaml`) and exits.
3. Each Java service applies its own changelogs on startup. Numbering is global: 001–041 plus the 1xx catalog seeds (040 of operations-service and 041 of notification-service are from 2026-10-21 and were applied on the server on 2026-10-22). In `develop` of 2026-10-23, 042 (operations-service, assignment notes) and 043 (booking-service, loyalty points per booking line) were added; they ship with `v1.1.0` together with payment-service's 021 (one promotion per booking); the next free one is 044.
4. Hibernate only validates that the model matches the tables (`ddl-auto: validate`); it never alters tables.

Production **does not load test data**: only catalogs (roles, services, statuses, schedule and business data).

4.2.2 Verification of Schema, Index and User Integrity

```
P=$(grep '^DB_PASSWORD=' .env.production | cut -d= -f2-)
q(){ sudo docker exec -e SQLCMDPASSWORD="$P" lavarapido-sqlserver /opt/mssql-tools18/bin/sqlcmd \
     -S localhost -U sa -C -d LavaRapido -Q "SET NOCOUNT ON; $1"; }
q "SELECT s.name AS esquema, COUNT(t.name) AS tablas FROM sys.schemas s JOIN sys.tables t ON t.schema_id=s.schema_id
GROUP BY s.name"
for s in SECURITY CUSTOMER BOOKING OPERATIONS NOTIFICATION PAYMENT; do
  q "SELECT '$s' AS servicio, COUNT(*) AS changesets, SUM(CASE WHEN EXECTYPE IN ('EXECUTED','MARK_RAN') THEN 1 ELSE 0
END) AS ok FROM DATABASECHANGELOG_$s"
done
q "SELECT COUNT(*) AS indices FROM sys.indexes WHERE object_id IN (SELECT object_id FROM sys.tables)"
q "SELECT u.username, r.code FROM security.app_user u JOIN security.user_role ur ON ur.user_id=u.user_id JOIN
security.role r ON r.role_id=ur.role_id WHERE r.code='ADMIN'"
```

Expected result:

- the 9 schemas with their tables;
- in each service, `ok` equal to `changesets` (2026-10-16: 143 of 143; 2026-10-19: 144 of 144, with 039); since 2026-10-22, 040 also appears in `DATABASECHANGELOG_OPERATIONS` and 041 in `DATABASECHANGELOG_NOTIFICATION`;
- the initial administrator created (security log: `First administrator created`).

4.3 Application Stack Deployment and Orchestration

4.3.1 Loading the docker-compose.yml File in Portainer

The stack is defined with two files versioned in `lavarapido-infra`:

- **`docker-compose.yml`:** the base, the same as in development.
- **`docker-compose.prod.yml`:** overrides the production settings: no internal ports, `prod` profile, memory limits, Caddy, web and CORS.

Current deployment (CLI):

```
# 1. Download the 9 repositories (and prepare Docker and swap the first time)
curl -fsSL https://raw.githubusercontent.com/BrayanCABRERA123/lavarapido-infra/main/deploy/setup-server.sh -o setup-
server.sh
bash setup-server.sh
```

The script is downloaded from `main` (version `v1.0.0`). In that version it clones and updates the `develop` branch of each repository (variable `BRANCH`). The one in `develop` (running on the server since 2026-10-24) clones `main` by default, except the repositories in `DEVELOP_ONLY` (today payment-service; since 2026-10-24 its `main` already has `v1.0.0` with its whole `develop`, so it can be removed from the list), and never downgrades: if a repository already pulled has something the requested branch does not contain, it stops without touching it. Until `v1.1.0` is released, use `BRANCH=develop bash setup-server.sh`.

With Portainer: Stacks → Add stack → *Repository*. The images are built from 9 sibling repositories, so the recommended flow is:

1. build the images with the CLI (step 4.3.3);
2. use Portainer to **view, restart, check logs and resource usage** of the containers of the `lavarapido` stack, which appears as a "limited" stack because it was created outside Portainer.

4.3.2 Injecting the Stack's Environment Variables

```
cd ~/lavarapido/lavarapido-infra
cp .env.production.example .env.production
chmod 600 .env.production
nano .env.production        # fill in the values from section 3.3 (keys generated with openssl)
```

In Portainer, the same variables are loaded in *Environment variables → Load variables from .env file*.

4.3.3 Service Deployment and Container Image Updates

First deployment:

```
cd ~/lavarapido/lavarapido-infra
sudo docker compose -f docker-compose.yml -f docker-compose.prod.yml \
  --env-file .env.production --profile app up -d --build
```

Later updates: first push the changes to GitHub and then (the script backs up the database before touching anything and, if the backup fails, does not update):

```
bash ~/lavarapido/lavarapido-infra/deploy/update-server.sh                # everything
bash ~/lavarapido/lavarapido-infra/deploy/update-server.sh web            # only the web
bash ~/lavarapido/lavarapido-infra/deploy/update-server.sh security-service notification-service
```

Until `v1.1.0` is released, manual updates are prefixed with `BRANCH=develop` (`BRANCH=develop bash …/update-server.sh`); without it, the protection in `setup-server.sh` stops. If everything is healthy, the script deletes unused images (the current one and `:prev` are kept) and build cache older than 7 days.

From GitHub, without SSH: with the machine on, in the `lavarapido-infra` repository → *Actions* → *Desplegar en Azure* → *Run workflow*. The button exists since 2026-10-23, when `deploy.yml` reached `main` (the default branch) with version `v1.0.0`; in that version the workflow calls `update-server.sh`, which deploys `develop`. In *Servicios*, leave it empty to update everything or write the names (`web payment-service`). At the end, the workflow checks that the web and the API answer 200. It uses its own GitHub key (secret `AZURE_SSH_KEY`) that on the server can only run `deploy/ci-deploy.sh`, which validates the names and does not let two updates run at the same time. **As of 2026-10-24 the workflow has never run** (that day's deployment was done over SSH) and the repository has no secrets, so it would fail until `AZURE_SSH_KEY` is created and its line installed in the server's `authorized_keys`; meanwhile, the SSH option above is used.

Continuous integration: since 2026-10-19 the 10 repositories have `.github/workflows/ci.yml`, which runs on every push or pull request to `develop` and `main`, and the result appears as ✓ or ✗ next to each commit. As of 2026-10-24, the latest `develop` run is green in all 10 (1,192 tests):

Repository	What it checks
Java services and api-gateway	`./mvnw verify`: builds and runs the tests (Java 21)
payment-service	`dotnet build` and `dotnet test` (.NET 8; integration tests are skipped without SQL Server)
Front-end-proyecto-web	Production build and `ng test`
LavadoVehicular-Movil	`tsc --noEmit` and `jest` (`jest-expo`, 25 suites and 73 tests). The APK is still built with EAS; `cd.yml` publishes the over-the-air update when it reaches `main` (4.3.4)
lavarapido-infra	Syntax and `shellcheck` of `deploy/*.sh`, `docker compose config`, `actionlint` of the workflows, and syntax of `e2e/` and `e2e/remote/`. The remote battery is run by hand with `e2e.yml` (not yet run against the server)

4.3.4 Continuous Deployment (Active Since 2026-10-26)

Continuous deployment has been merged into `develop` since 2026-10-24: `lavarapido-infra` (`006b5b0`) and the **Pedir despliegue** (request deployment) job in the 8 repositories deployed to Azure (web, api-gateway, security, customer, booking, operations, notification, and payment), plus `cd.yml` in the mobile app. It reached `main` with `v1.1.0` and, with the secrets created, **it ran on its own for the first time on the night of 2026-10-26** (section 1.3.12). This is how it works:

1. When an approved PR enters `main`, the repository's `ci.yml` runs the CI and, if it passes, its **Pedir despliegue** job launches the **Desplegar en Azure** workflow of `lavarapido-infra` with `automatic=true` (secret `DEPLOY_DISPATCH_TOKEN`: a *fine-grained* token with access only to `lavarapido-infra` and *Actions: Read and write* permission). A merge into `main` of `lavarapido-infra` also deploys. The ✓ of **Pedir despliegue** only says the request arrived; the result is in *Actions* → *Desplegar en Azure* of `lavarapido-infra`.
2. The workflow logs in over SSH (`AZURE_SSH_KEY`), backs up the database, pulls `main` of every repository (since infra *v1.1.1* payment-service too, which used to stay on `develop` through `DEVELOP_ONLY`), and rebuilds what changed, first keeping the running version as `:prev`. If another update is already running, for example the one at machine startup, it waits up to 20 minutes and then updates. Then it waits up to 8 minutes for **all** containers to be healthy: if one does not start, the workflow ends red and says which. If the `Caddyfile` changed, it restarts Caddy. At the end it checks that the web and the API answer 200 and, if not, shows what the API answered.
3. If the machine is **off** (neither SSH nor the site answers), the automatic workflow leaves a notice and does not fail; if the site answers but SSH does not, it ends red. When it is turned on, `deploy/boot-update.sh` (an `@reboot` task, installed on the server since 2026-10-24) pulls what is new in `main` and updates if something is not deployed, comparing with `~/lavarapido/backups/deployed-heads`, which is only written when a complete update ends well. While the server is on `develop` (until `v1.1.0`), the protection against downgrading stops that update without touching anything. The result is in `~/lavarapido/backups/deploy.log`.
4. **Mobile app (OTA):** it is not deployed to Azure. Its `cd.yml` runs on each push to `main` (not for changes to `.md` or `docs/` only) or by hand, but only publishes from `main`: it repeats the CI (`tsc` and `jest`), compares the native fingerprint of the code with that of the published APK and, if it matches, runs `eas update --channel preview`; if it changed, it fails asking for a new APK (`eas build` and `publish:apk`, still manual). It uses the `EXPO_TOKEN` secret. Since MOV-08 changed `eas.json`, the mobile app's first step to `main` will ask for a new APK.

How it was activated (AC-90 of the Corrective Actions document; the secrets exist since 2026-10-26):

Repository	Secret	Created by
`lavarapido-infra`	`AZURE_SSH_KEY` (and its `restrict,command="bash …/deploy/ci-deploy.sh"` line in the server's `~/.ssh/authorized_keys`)	Brayan Patiño
web, api-gateway, security, operations, and notification	`DEPLOY_DISPATCH_TOKEN`	Brayan Patiño
customer and booking	`DEPLOY_DISPATCH_TOKEN` (the same token)	Juan Camilo Guilombo or someone with the Admin role
payment	`DEPLOY_DISPATCH_TOKEN`	Merari Urbano or someone with the Admin role
LavadoVehicular-Movil	`EXPO_TOKEN` (the project's Expo account)	Merari Urbano or someone with the Admin role

If `DEPLOY_DISPATCH_TOKEN` is missing, that repository's **Pedir despliegue** job ends red even if the CI passes. **Order of the `v1.1.0` release:* *first the web, the services, and payment (their deployment request fails with *422* while `deploy.yml` in infra's `main` does not accept `automatic`, with no harm) and `lavarapido-infra` last; then payment is removed from `DEVELOP_ONLY`. It is checked with `crontab -l | grep boot-update` and `git -C ~/lavarapido/Front-end-proyecto-web branch --show-current` (it must say `main`).

Publishing the mobile app, from the PC, in the app folder:

```
eas build -p android --profile preview          # new APK (native changes)
npm run publish:apk -- "what changed"           # publishes it in /app/ and notifies by push
npm run publish:update -- "what changed"        # screen-only changes: over-the-internet update
```

4.4 Routing, Domain and SSL/TLS Certificate Configuration

1. **Domain:** Azure → public IP `vehiclecarwash-vm-ip` → Configuration → *DNS name label* = `lavarapidovehicular`. The result is `lavarapidovehicular.northcentralus.cloudapp.azure.com`.
2. **Variables:** `.env.production`: `` `SITE_ADDRESS=lavarapidovehicular.northcentralus.cloudapp.azure.com` `PUBLIC_URL=https://lavarapidovehicular.northcentralus.cloudapp.azure.com`
3. **Apply:** `update-server.sh caddy security-service notification-service api-gateway`. Caddy requests the certificate from Let's Encrypt through HTTP-01; port 80 must be open.
4. **Verify the certificate:** ``bash echo | openssl s_client -connect lavarapidovehicular.northcentralus.cloudapp.azure.com:443 \ -servername lavarapidovehicular.northcentralus.cloudapp.azure.com 2>/dev/null | openssl x509 -noout -issuer -enddate # issuer=C=US, O=Let's Encrypt · notAfter=Jan 14 2027`
5. **Mobile app:** `EXPO_PUBLIC_API_URL` in `eas.json` = `https://lavarapidovehicular.northcentralus.cloudapp.azure.com/api/v1`, and the APK is built.

5. Verification, Testing and Evidence Plan (Smoke Tests)

5.1 Service Verification Links and Endpoints Matrix

5.1.1 Access to the Administration Panel (Portainer)

Link	Expected	Status
`https://localhost:9443` (through SSH tunnel)	Stack `lavarapido`: 11 containers *running/healthy* and 2 init containers in *exited (0)*	✅ (see 5.4.1)
`https://64.236.160.129:9443` (from the internet)	No connection	✅ closed

5.1.2 Public Web Interface (Angular SPA)

Link	Expected	Result (2026-10-19)
`https://<domain>/`	Home page with the "Download app" section	✅ 200
`https://<domain>/client`, `/admin`	Angular routes (load `index.html`)	✅ 200
`http://<domain>/`	Redirect to HTTPS	✅ 308
`https://<domain>/app/version.json`	Published app version	✅ 200 (`versionCode` 5, version 1.1.0)
`https://<domain>/app/LavaRapidoVehicular.apk`	APK download (`application/vnd.android.package-archive`)	✅ 200

5.1.3 Backend Health Endpoints (/actuator/health)

The health endpoints **are not published to the internet**: `/actuator` is not under `/api`. They are checked inside the server:

```
for s in security-service:3001 customer-service:3002 booking-service:3003 operations-service:3004 notification-
service:3006 api-gateway:8080; do
  sudo docker run --rm --network lavarapido_default curlimages/curl:8.10.1 -s "http://$s/actuator/health"; echo "  ←
$s"
done
sudo docker compose -f docker-compose.yml -f docker-compose.prod.yml --env-file .env.production --profile app ps
```

Expected: `{"status":"UP"}` in each Java service, and every container `Up (healthy)`. `db-init` and `payment-migrate` must end in `Exited (0)`.

5.1.4 Interactive API Documentation (Swagger UI / OpenAPI)

- **In production it is disabled on purpose** (`prod` profile): the API structure is not exposed.
- It is consulted in **development**, with each service in the `dev` profile: `http://localhost:3001/swagger-ui.html`, and likewise for each port.
- The OpenAPI specification is versioned in `vehicle-w-docs`.

5.1.5 Direct TCP Connectivity to the Database Engine

SQL Server **does not have port 1433 published**. Connectivity is tested from the internal network:

```
P=$(grep '^DB_PASSWORD=' .env.production | cut -d= -f2-)
sudo docker exec -e SQLCMDPASSWORD="$P" lavarapido-sqlserver /opt/mssql-tools18/bin/sqlcmd \
  -S localhost -U sa -C -Q "SELECT @@VERSION, DB_NAME(DB_ID('LavaRapido'))"
nc -zv 64.236.160.129 1433    # from outside: must FAIL (port closed)
```

5.2 Functional Smoke Tests

5.2.1 Authentication Flow and Token Issuance

Test	Expected	Result
`POST /api/v1/auth/login` with an empty body	400 (validation)	✅ 400
Login with invalid credentials	401 `INVALID_CREDENTIALS`	✅ 401
Login of the initial administrator from the web	JWT token with the `ADMIN` role and access to `/admin`	✅
Request from a foreign origin (CORS)	403	✅ 403
Internal route without key (`/internal/...`)	401; unreachable from the internet	✅ 401 / 405

5.2.2 Data Read and Query Tests

Test	Result
`GET /api/v1/catalog/services` (public catalog)	✅ 200 with the services
`GET /api/v1/schedule/business-hours`	✅ 200
`GET /api/v1/establishment`	✅ 200

5.2.3 Write and Transactional Persistence Tests

Automated end-to-end test: `lavarapido-infra/e2e/full-flow.test.js` (`npm run flow`). It runs only locally, against the containers of `docker compose --profile app` (it reads the local `.env` and uses `docker exec`); the GitHub CI only checks its syntax (`node --check`) and does not run it against the server. It goes through the three roles via the gateway:

#	Step	Result
1	The client books choosing an operator → automatic assignment → notices	✅
2	The client reports the payment with a receipt → it stays under review	✅
3	The administrator approves the payment → the client sees it approved and earns points	✅
4	The operator starts and finishes the service → notices to the client	✅
5	The client rates → the operator sees the rating	✅

- **When it ends:** the test deletes everything it created and restores the client's points.
- **Also verified in production:**
- real email delivery (recovery code through Gmail);
- real push to the phone;
- the push switch (off → inbox only; on → push).

5.2.4 Security Headers Check

Since 2026-10-20 Caddy adds the security headers to every response (web, API and downloads), and to the web a content security policy (CSP) that only allows the project's own code plus Google Fonts and the jsDelivr flags.

```
curl -s -I https://lavarapidovehicular.northcentralus.cloudapp.azure.com/ | grep -iE "strict|content-security|x-frame|x-content|referrer|permissions|server"
```

Test	Expected	Result (2026-10-20)
Headers on the web, `/api` and `/app`	HSTS, `nosniff`, `X-Frame-Options: SAMEORIGIN`, `Referrer-Policy`, `Permissions-Policy`; CSP on the web	✅ No duplicated headers
`Server` header	Not present (it used to show `nginx/1.27.5`)	✅
Web with the CSP on (home, login, sign-up, password recovery)	Loads fully: styles, fonts, flags, QR	✅ 0 blocks and 0 console errors
Load an external script from the browser console (simulated attack)	The browser blocks it: *Refused to load the script … Content Security Policy*	✅ Blocked

5.3 Log Inspection and Resource Monitoring

5.3.1 Review of Startup Traces Without Critical Exceptions

```
C="sudo docker compose -f docker-compose.yml -f docker-compose.prod.yml --env-file .env.production --profile app"
$C logs --since 10m | grep -iE "ERROR|Exception|APPLICATION FAILED" | head
for s in security-service customer-service booking-service operations-service notification-service api-gateway; do
  sudo docker logs lavarapido-$s 2>&1 | grep -oE "Started [A-Za-z]+ in [0-9.]+ seconds" | tail -1
done
sudo docker logs lavarapido-payment-service 2>&1 | grep "Now listening"
```

Expected:

- each service with its `Started ... in N seconds`;
- payment with `Now listening on: http://0.0.0.0:3005`;
- no `APPLICATION FAILED TO START`.

5.3.2 CPU and RAM Usage Monitoring in Containers

```
sudo docker stats --no-stream --format "table {{.Name}}\t{{.CPUPerc}}\t{{.MemUsage}}"
free -h
```

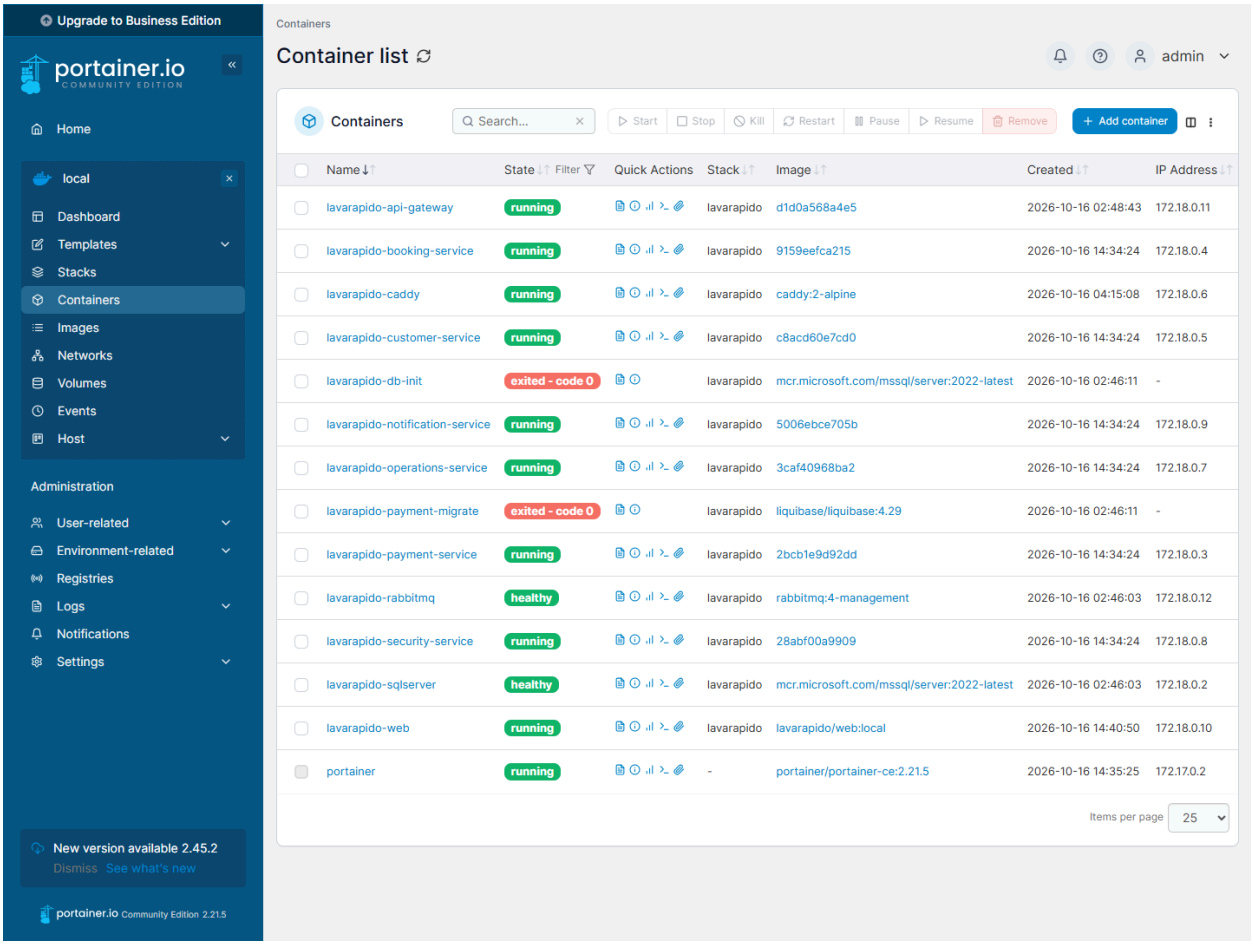
Measurement on 2026-10-19, with the system idle and the machine just started:

Container	Memory
security-service	311 MB / 512 MB
notification-service	326 MB / 512 MB
operations-service	327 MB / 512 MB
booking-service	341 MB / 512 MB
customer-service	292 MB / 512 MB
api-gateway	176 MB / 512 MB
payment-service	78 MB / 384 MB
sqlserver	777 MB
rabbitmq	148 MB
web / caddy	12 MB / 60 MB
portainer	61 MB
Machine total	**3.3 GB of 7.7 GB · swap unused**

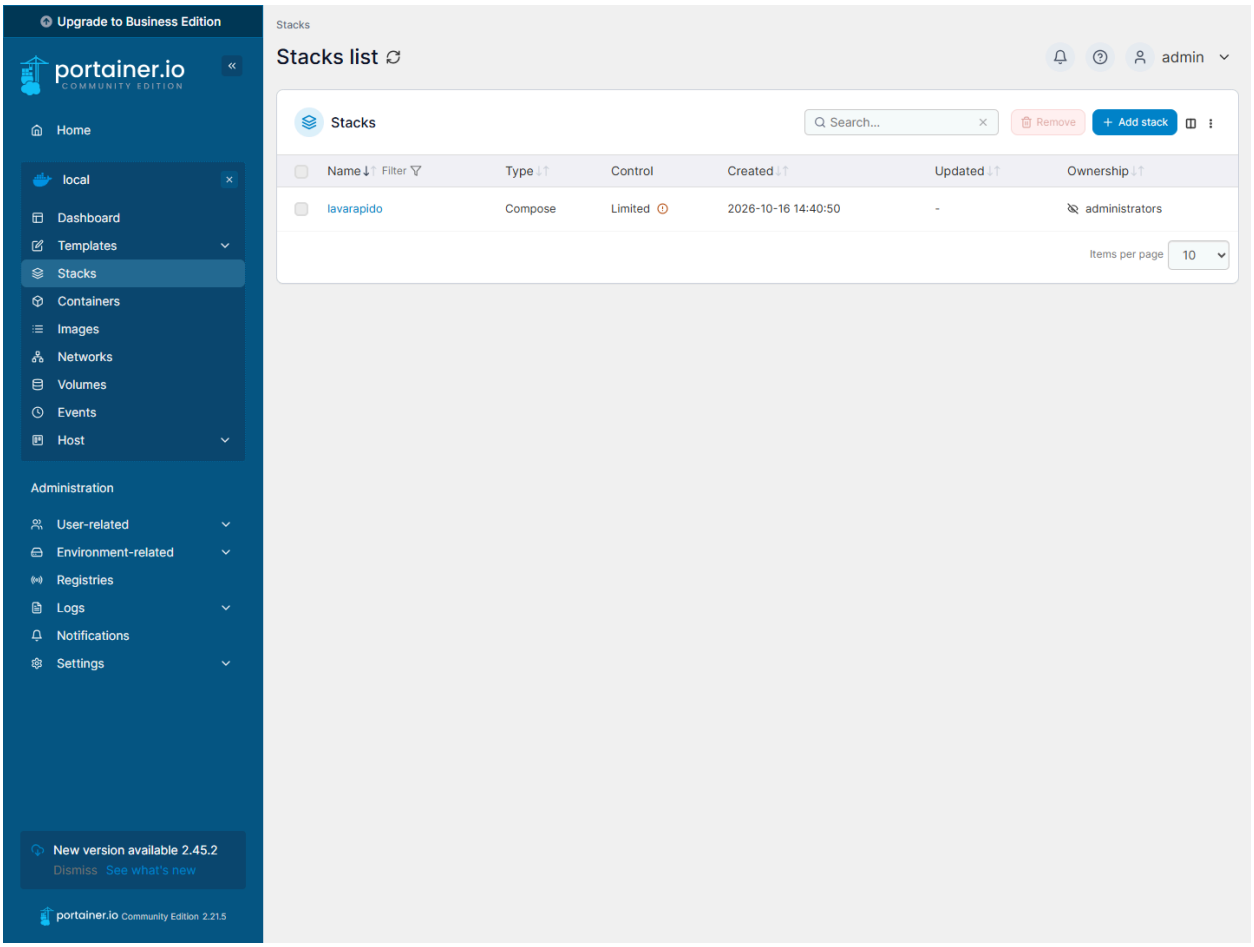
5.4 Visual Evidence Record (Screenshots)

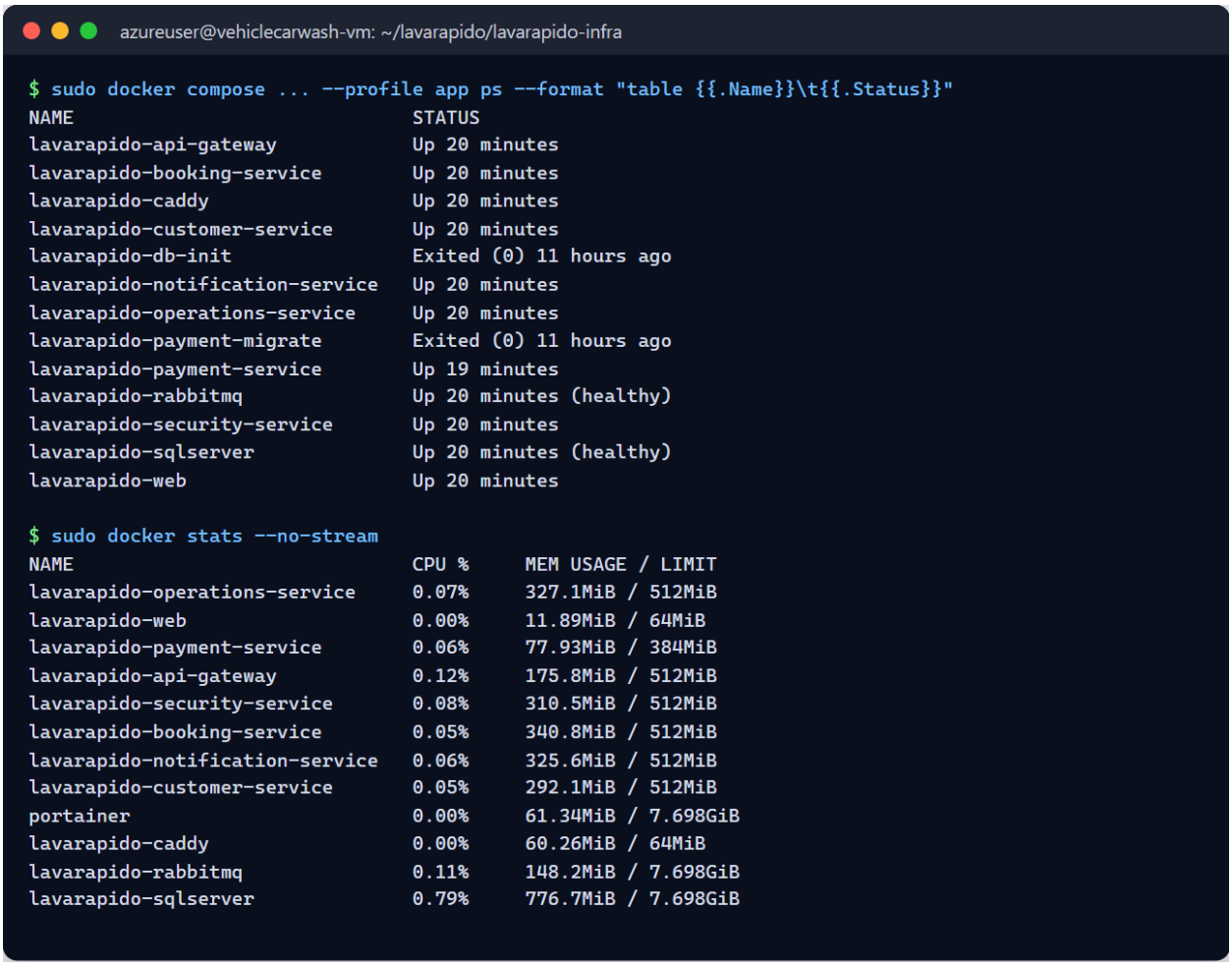
> The terminal and web screenshots were taken on 2026-10-19 on the production server, with the version deployed that day at 03:20; the terminal ones show the real output of the commands. The Portainer ones (5.4.1) are from 2026-10-16, because the panel requires logging in and only the DevOps owner uses it.

5.4.1 Screenshot of the Portainer Panel With Containers in Running State



The 11 services in running / healthy*; `db-init` and `payment-migrate` end in* exited - code 0 *because they run only once (create the database and apply the payment migrations).*





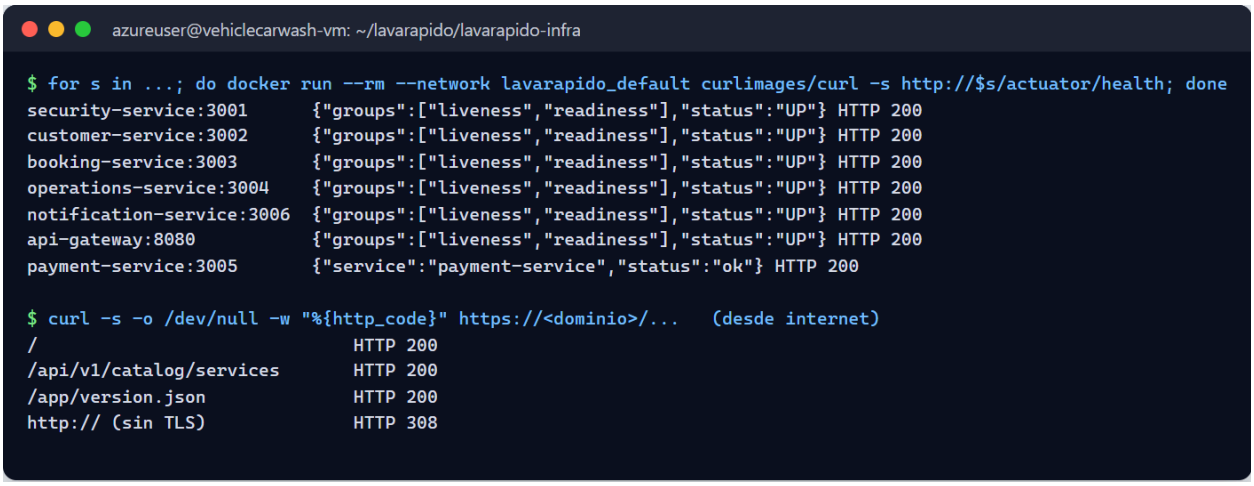
Status taken from the terminal on 2026-10-19: the 11 containers of the system Up (SQL Server and RabbitMQ healthy*), `db-init` and `payment-migrate` in* Exited (0)*, and the memory usage of each container.*

5.4.2 Screenshot of the Backend Services Log Viewer



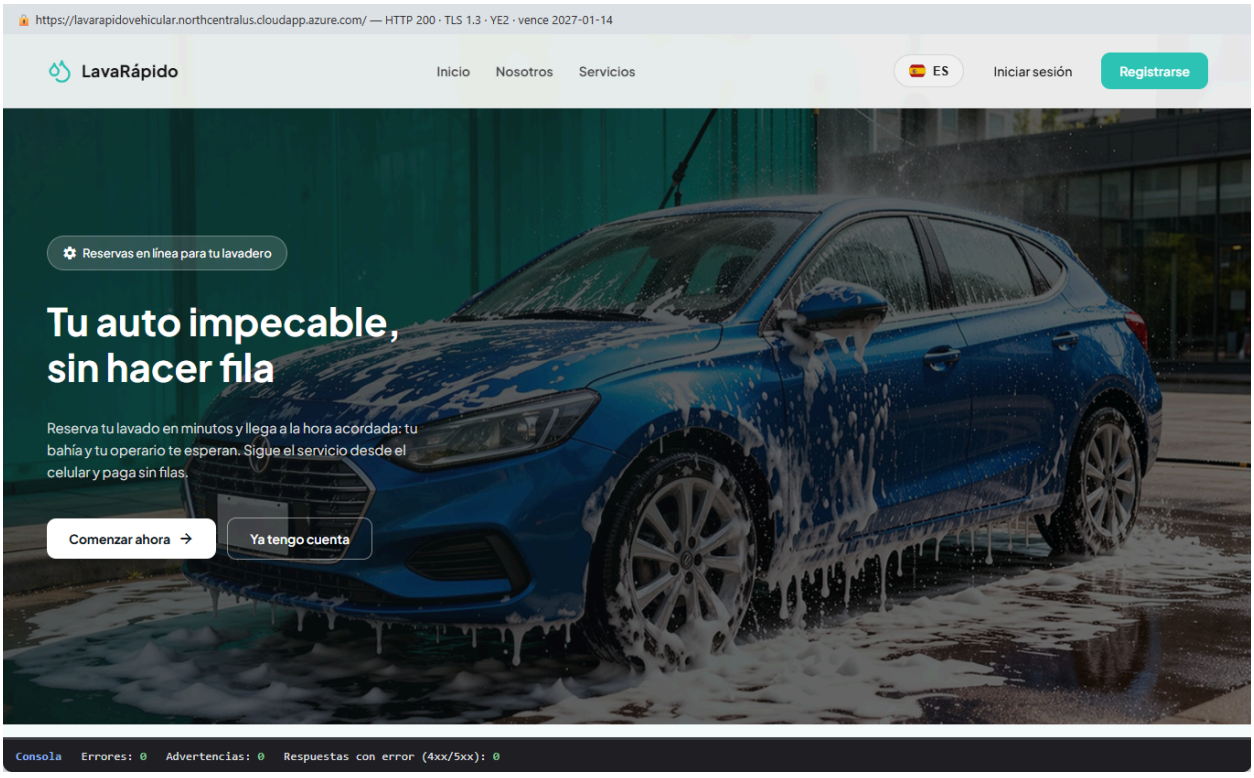
Each Java service shows Started … in N seconds *(between 45 and 91 seconds with the machine just started) and payment* Now listening on :3005*. There are 0 error traces in the last 15 minutes.*

5.4.3 Screenshot of the HTTP 200 Response of the Health Checks



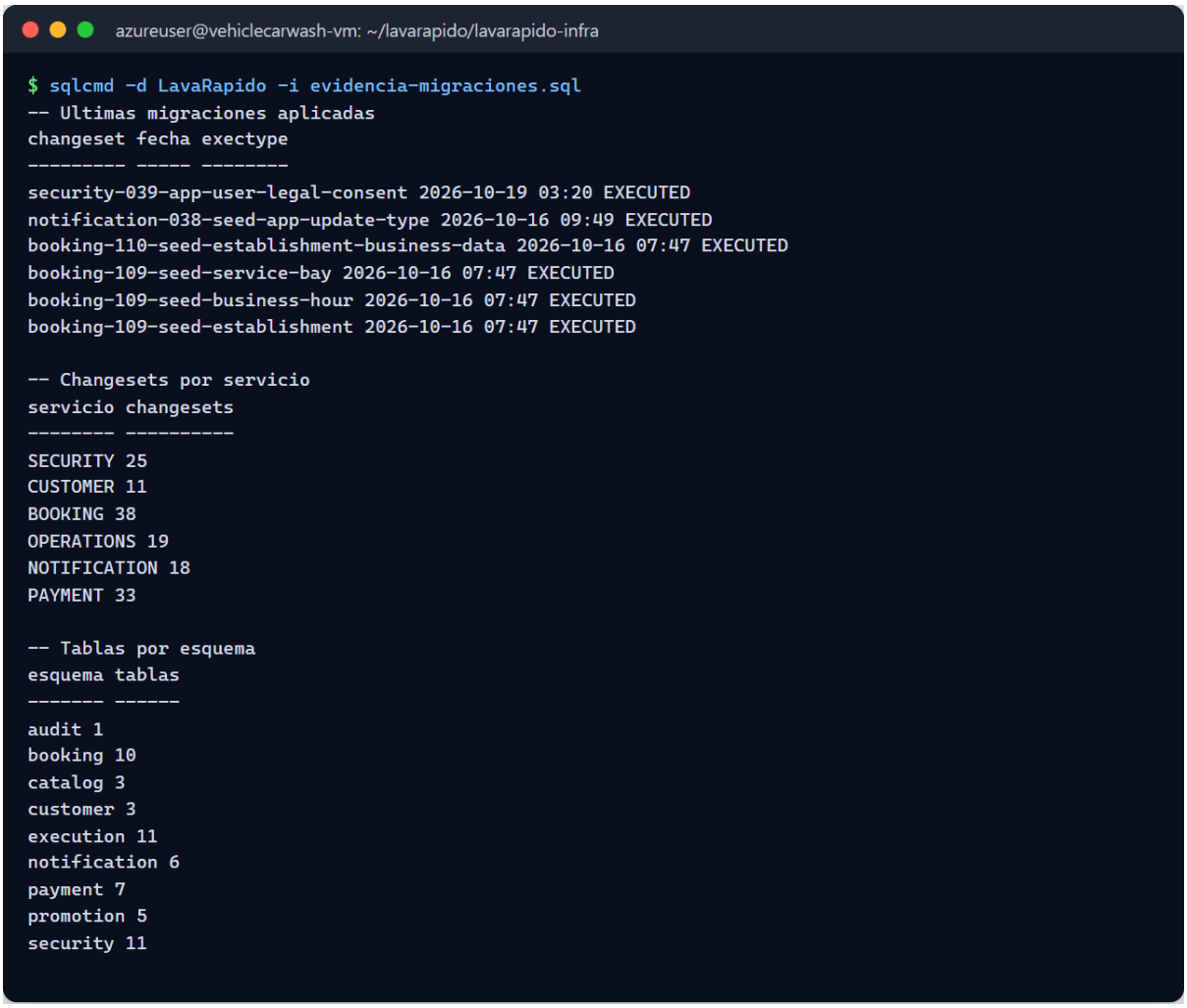
`/actuator/health` answers `UP` (HTTP 200) in the 6 Java services from the internal network, and payment-service answers `ok`. From the internet, the web, the API, and `/app/version.json` answer 200, and HTTP redirects to HTTPS with 308.

5.4.4 Screenshot of the Web Interface With a Clean DevTools Console



Home page in production over HTTPS (TLS 1.3, Let's Encrypt certificate valid until 2027-01-14). On load: 0 errors and 0 warnings in the console, and 0 error responses.

5.4.5 Screenshot of the Successful Migration Record in the Database



Migration 039 (acceptance of the legal documents) was applied on 2026-10-19; there are 144 `EXECUTED` changesets in the 6 histories and 57 tables in the 9 schemas.

Related documents:

- `lavarapido-infra/DEPLOY.md`: short deployment and update guide.
- `lavarapido-infra/e2e/README.md`: tests between services.
- `vehicle-w-docs`: architecture (ADR), OpenAPI and events.